

Customer Complaint Process Mining & BPMN Redesign

Data Set: [Complaints filed by customers \(BPI Challenge 2016\)](#)

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Introduction

This assignment applies process mining techniques to analyze a complaint-handling process using the BPI Challenge 2016 dataset from the Dutch Employee Insurance Agency (UWV). Based on an event log containing customer complaint activities, the current process is analyzed to identify key process issues. Building on these findings, redesign heuristics are applied to propose an improved process model that addresses the identified weaknesses in terms of time and quality.

Issue 1: Lack of Process Standardization and High Variability

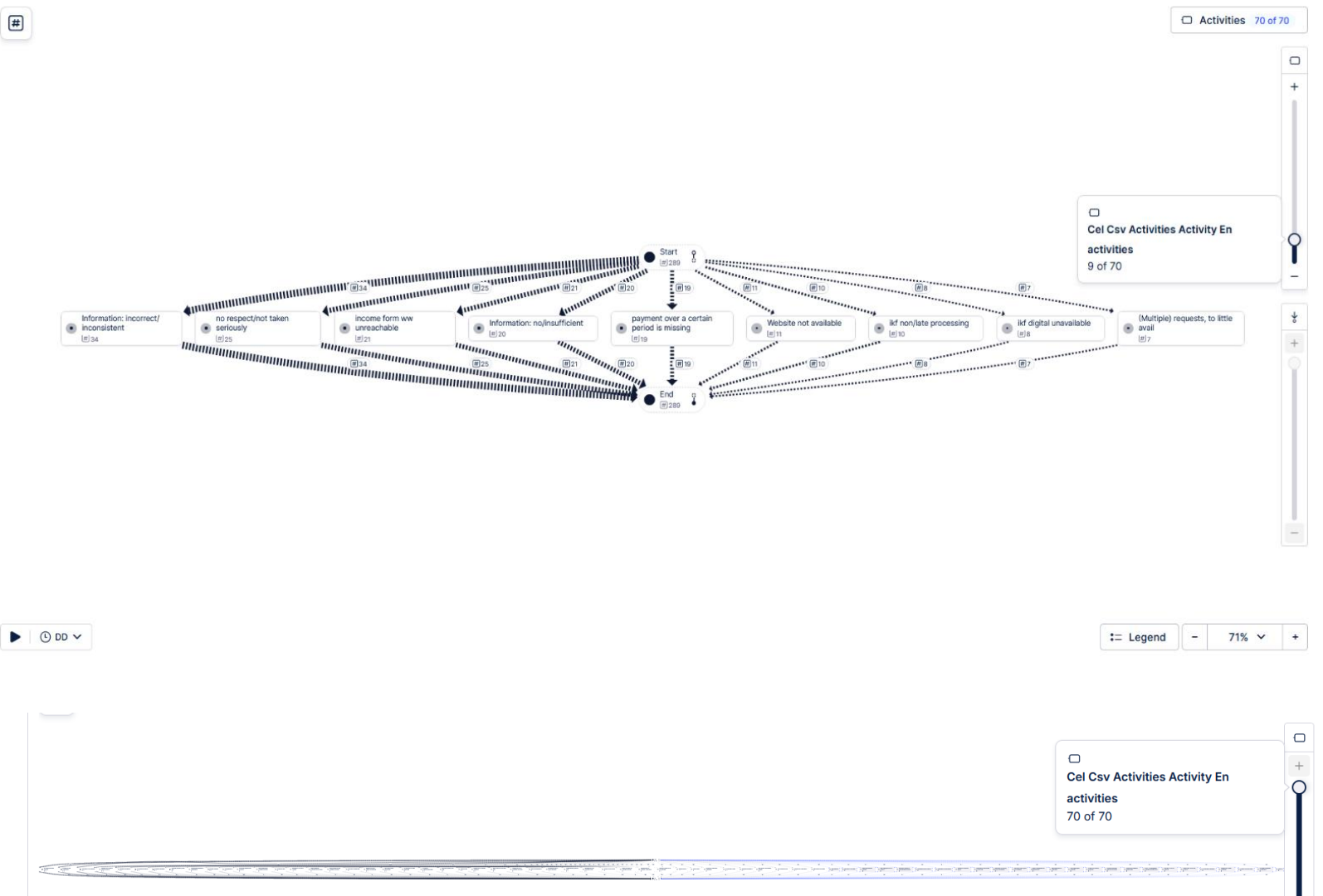


Fig 1: The Process Explorer shows a highly parallel and weakly structured process.

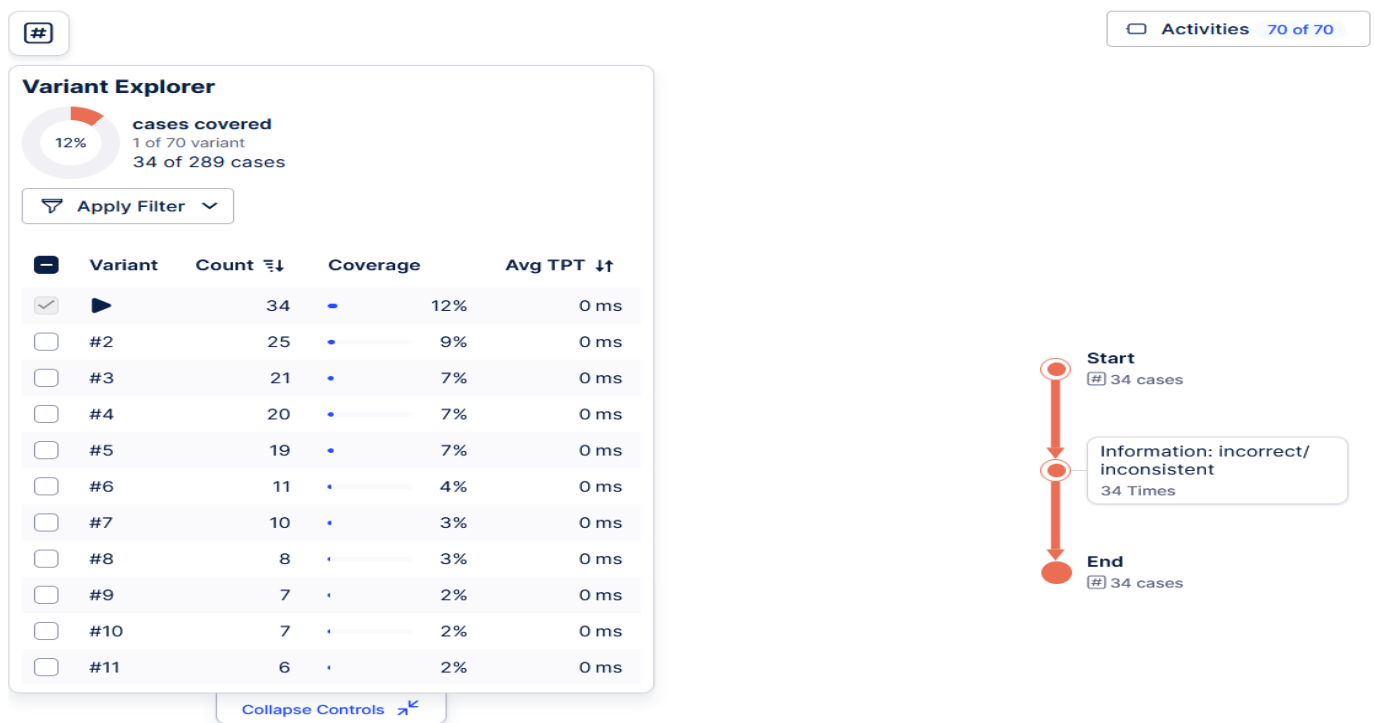


Fig 2: Variant explorer shows many variants and Low coverage per variant.

The Process Explorer and Variant Explorer show a large number of variants with very low coverage per variant, meaning that no dominant or standardized complaint-handling path exists. This indicates that complaints are handled in a highly case-by-case manner rather than following a common control flow. Such high process variability increases complexity, reduces transparency, and makes it difficult to manage performance consistently. The observed variant explosion suggests a lack of standard operating procedures and limited guidance for handling recurring complaint types.

Issue 2: Lack of End-to-End Process Visibility and Unobservable Waiting Times

Assignment View

Case Id	# of Activities	Throughput Time	First Activity	First Activity Timestamp	Last Activity	Last Activ	Case details: 99
99	1	-	not received ikf	11/03/15 00:00:00	not received ikf	11/03/15 00:00:00	Search
98	1	-	Information: no/insufficient	10/30/15 00:00:00	Information: no/insufficient	10/30/15 00:00:00	Activities
97	1	-	missed appointments	10/30/15 00:00:00	missed appointments	10/30/15 00:00:00	not received ikf
96	1	-	after contact kcc not called back	10/30/15 00:00:00	after contact kcc not called back	10/30/15 00:00:00	11/03/15 00:00:00
95	1	-	Continued payment does not follow	10/29/15 00:00:00	Continued payment does not follow	10/29/15 00:00:00	
94	1	-	payout Toeslagenwet not given/irregular	10/30/15 00:00:00	payout Toeslagenwet not given/irregular	10/30/15 00:00:00	
93	1	-	disruption: payment late	10/23/15 00:00:00	disruption: payment late	10/23/15 00:00:00	
92	1	-	Info unclear/incorrect	10/27/15 00:00:00	Info unclear/incorrect	10/27/15 00:00:00	
91	1	-	no internal communication/data transfer	10/23/15 00:00:00	no internal communication/data transfer	10/23/15 00:00:00	
90	1	-	missed appointments	10/20/15 00:00:00	missed appointments	10/20/15 00:00:00	
9	1	-	(Multiple) requests, to little avail	7/09/15 00:00:00	(Multiple) requests, to little avail	7/09/15 00:00:00	
89	1	-	payment over a certain period is missing	10/23/15 00:00:00	payment over a certain period is missing	10/23/15 00:00:00	
88	1	-	Information: no/insufficient	10/22/15 00:00:00	Information: no/insufficient	10/22/15 00:00:00	
87	1	-	recalculation of the benefit is not/too late	10/23/15 00:00:00	recalculation of the benefit is not/too late	10/23/15 00:00:00	
86	1	-	Information: incorrect/inconsistent	10/20/15 00:00:00	Information: incorrect/inconsistent	10/20/15 00:00:00	
85	1	-	change form is not processed, incorrect payment	10/16/15 00:00:00	change form is not processed, incorrect payment	10/16/15 00:00:00	
84	1	-	change form non/late processing	10/16/15 00:00:00	change form non/late processing	10/16/15 00:00:00	
83	1	-	ikf non/late processing	10/15/15 00:00:00	ikf non/late processing	10/15/15 00:00:00	
82	1	-	ikf non/late processing	10/13/15 00:00:00	ikf non/late processing	10/13/15 00:00:00	
81	1	-	not/hard to reach	10/13/15 00:00:00	not/hard to reach	10/13/15 00:00:00	
80	1	-	Information: incorrect/inconsistent	10/14/15 00:00:00	Information: incorrect/inconsistent	10/14/15 00:00:00	
8	1	-	Information: incorrect/inconsistent	7/09/15 00:00:00	Information: incorrect/inconsistent	7/09/15 00:00:00	
79	1	-	disruption: payment late	10/10/15 00:00:00	disruption: payment late	10/10/15 00:00:00	

Fig 3: Case Explorer

Due to each case containing only one recorded activity, end-to-end throughput times cannot be measured. This indicates a lack of process visibility and suggests that waiting times and internal handling occur outside the event log. Waiting time cannot be observed and the system does not track progress beyond the first contact. Due to this data limitation, the organization is unable to monitor process performance, identify bottlenecks, or systematically analyze time-related inefficiencies in complaint resolution.

Issue 3: Fragmented Case Handling and Rework Across Complaints

Assignment View

Case Id	# of Activities	Throughput Time	First Activity	First Activity Timestamp	Last Activity	Last Activity Timestamp
99	1	-	not received ikf	11/03/15 00:00:00	not received ikf	11/03/15 00:00:00
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97	1	-	missed appointments	10/30/15 00:00:00	missed appointments	10/30/15 00:00:00
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Fig 4: Case Explorer

The analysis indicates that complaints are not handled as continuous cases over time. Since each case consists of only one recorded activity, repeated customer contacts are likely registered as new, independent cases rather than being linked to an existing complaint. This fragmented handling leads to duplicated work, inconsistent responses, and reduced service quality from the customer's perspective. The lack of case continuity suggests that customer issues are not resolved within a single process instance, increasing rework for employees and negatively affecting customer satisfaction and overall process efficiency.

Redesign for Issue 1: Lack of Standardization

Heuristic used: Standardization

To reduce process variability, the redesigned process applies standardization by enforcing a fixed sequence of activities in the BPMN model. All complaints follow the same main flow from complaint registration, categorization by department, assignment by department, investigation, resolution, and closure. A single XOR gateway ("Simple complaint?") structures decision-making. This standardized backbone reduces variant explosion, shortens decision time through predefined routing, and ensures consistent handling and outcomes for similar complaint types.

Redesign for Issue 2: Lack of End-to-End Visibility

Heuristic used: Case-based work

End-to-end visibility is improved by redesigning the process as a case-based workflow. In the BPMN model, a complaint is represented as one persistent case that starts with complaint submission and ends with closure. All activities, including investigation and information requests, remain within the same process instance. This allows throughput times and waiting periods to be measured across the full lifecycle, improving transparency, monitoring, and coordination.

Redesign for Issue 3: Fragmented Handling and Rework

Heuristic used: Elimination of rework (Quality at the source)

Rework is reduced by ensuring that repeated customer interactions are handled within the same complaint case. In the BPMN model, requests for additional information and subsequent resolutions are modeled as part of the same process flow rather than creating new cases. This prevents duplicate work, avoids inconsistent responses, and improves customer experience by maintaining continuity and responsibility throughout the complaint resolution process.

Ideal BPMN Model

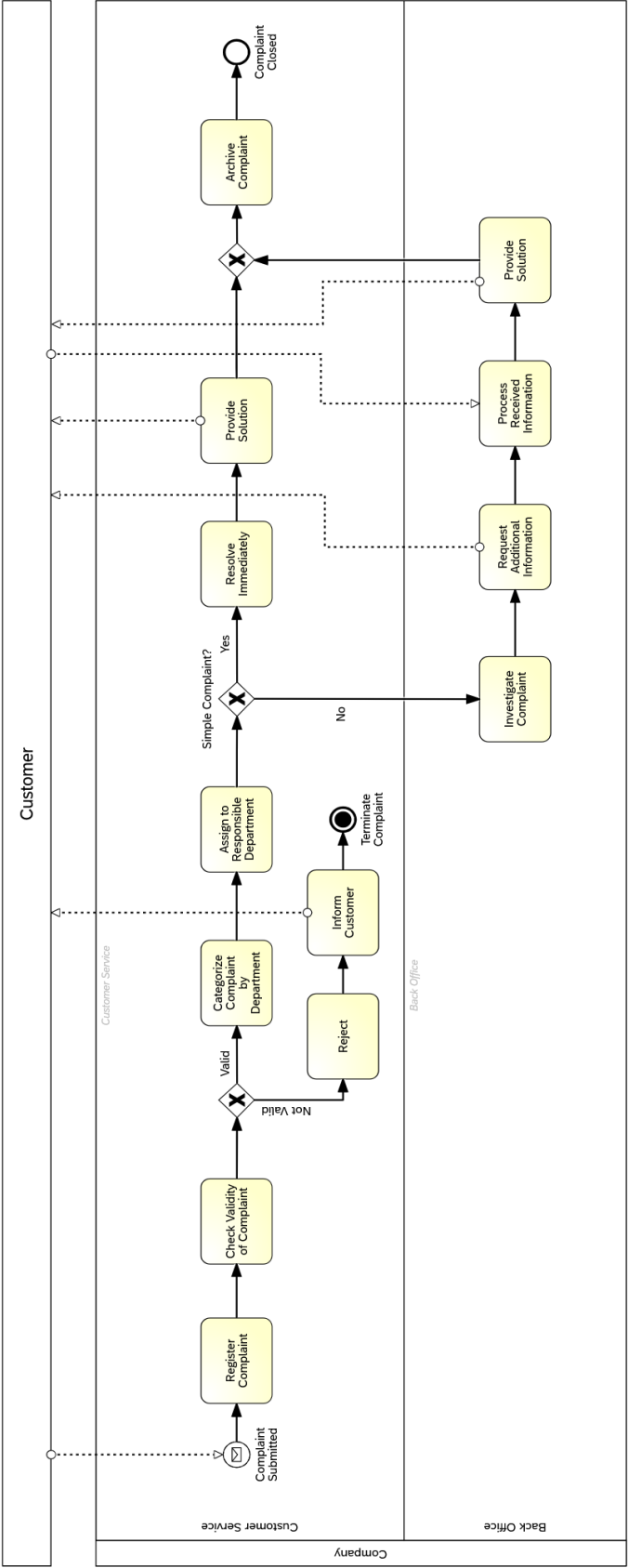


Fig 5: BPMN Model